OPEN AGENT HACKATHON 2026 · TRACK 2 — THE AGENT THAT CAN EXPLAIN WHY

RecallWhy

Agents that tell a retailer which recalled products are on its shelves, which lots, which stores, how many customers to call — **and why they believe it.**

Zetaris federated SQL · Meterless H-MEM + World Model · NVIDIA Nemotron · built in Cursor

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github.com/mikey92/recallwhy

1 · PROBLEM AND USERS

517 federal recall notices since July. Each one is a manual cross-check.

A regional retailer's **product-safety / QA manager** reads every CPSC and FDA notice and must answer:

- Do we sell this — or something that only looks like it?
- Which lots? Which stores still hold them?
- How many customers bought it since the recall began?
- Who supplied it?

184 + 333

CPSC recalls and FDA food enforcement reports, 1 Jul – 21 Oct 2026, read live by RecallWhy

The answers live in three places that don't talk to each other: the **recall feeds**, the **ERP** and the **supplier portal**. A wrong "not ours" leaves a contaminated lot on the shelf; a wrong "ours" pulls good stock.

2 · SOLUTION

Four agents investigate every recall and return a verdict with its evidence.

Scout

Reads both live feeds and the item master through Zetaris; one federated UPC sweep; opens a case per recall that touches the catalog.

Investigator

UPC / model rules, Nemotron judge for text matches, lot codes, prior decisions from H-MEM, exposure by store and lot with federated SQL.

Auditor

Can block. Re-checks the live notice, quotes and lots verbatim, re-derives every number with its own SQL, checks each brief sentence. ≤2 revisions, then a human.

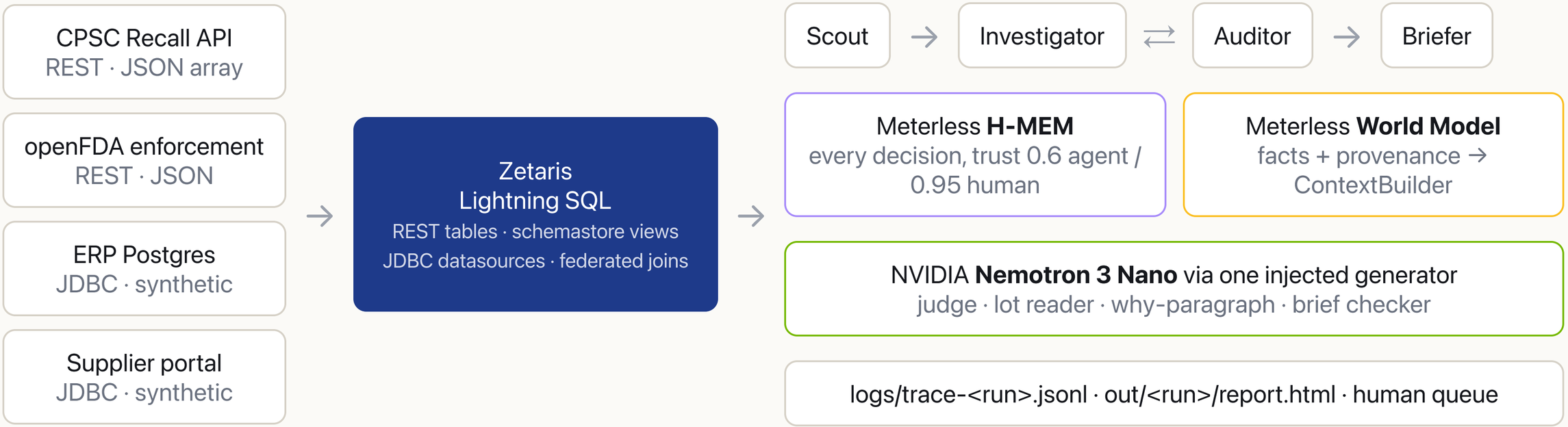
Briefer

Writes only from the World Model context: numbered evidence, each with the SQL or model call behind it.

ACT NOW · NOTIFY · NO EXPOSURE · NOT OURS · NEEDS HUMAN

3 · ARCHITECTURE

Every agent reads data only through Zetaris.



4 · STACK

Small, typed, runnable from a clean clone.

LAYER	WHAT
Data access	Zetaris Lightning SQL over its REST API (`/api/proxy/sql-editor/sqls/run-query`)
Agents	TypeScript on Node 22, no agent framework — four modules, one orchestrator, one trace
Memory & state	Meterless H-MEM and World Model reference engines (Apache-2.0), mounted in `src/engines/`
Model	NVIDIA **nemotron-3-nano** (30B-A3B) via Ollama; any OpenAI-compatible endpoint by env var
Retailer data	Postgres 16 in Docker — synthetic ERP + supplier portal (seeded generator)
No-credential mode	`SAMPLE_MODE=true` : same agents and SQL on an in-process DuckDB stand-in, labelled SAMPLE
Development	Cursor (agent scaffolding, generator, SQL layer, tests) · 37 tests · `npm run view` live trace viewer

5 · HOW ZETARIS AND METERLESS ARE USED

Load-bearing, not bolted on.

Zetaris

- 2 live Lightning REST tables → schemastore views
- 2 JDBC datasources (ERP, supplier portal)
- Scout: **REST × JDBC** join — live FDA notice digits ⋈ ERP UPCs
- Investigator: ship notices ⋈ suppliers ⋈ ERP item master
- Auditor: re-queries the live notice and re-aggregates every number itself
- Setup is code: `npm run setup:zetaris`

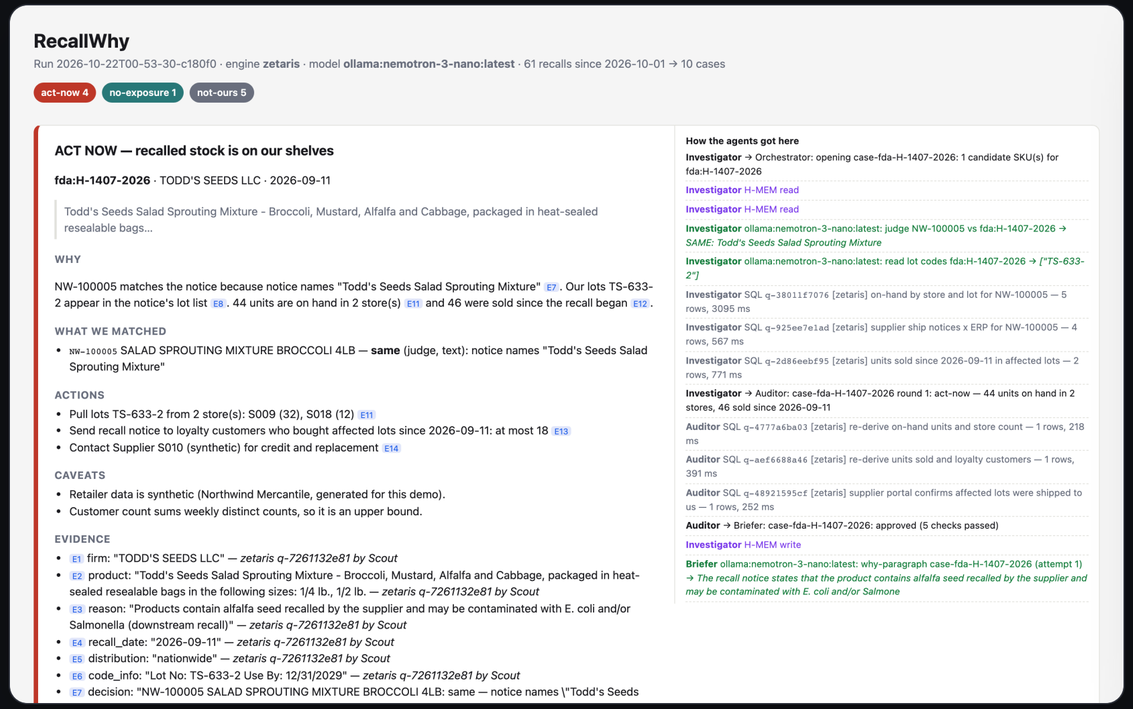
Meterless

- **H-MEM**: every adjudication stored with trust; human decisions (0.95) are followed on the next run; audited agent decisions are reused without a model call; trust ledger logs reads/writes
- **World Model**: recalls, cases, findings as facts; every write carries provenance (agent, SQL id, model call)
- **ContextBuilder** is the Briefer's only input — it cannot cite what the team didn't establish

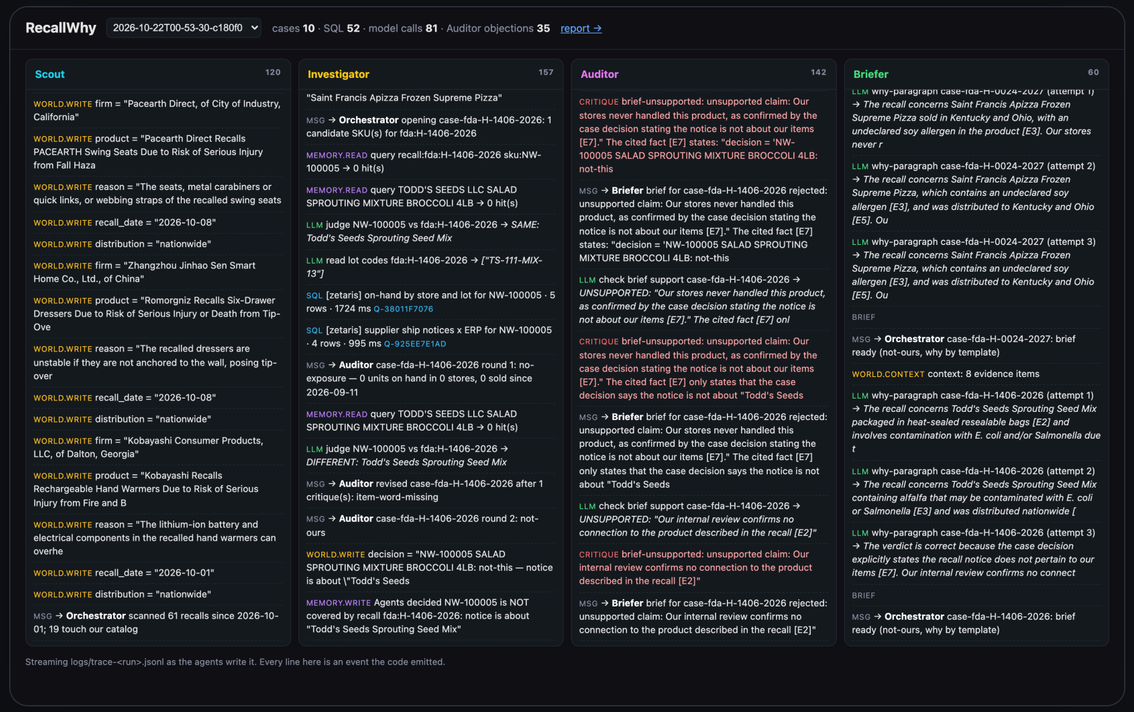
The Auditor doesn't trust anyone — including the model.

- **Re-derives every number in Zetaris** with differently written SQL; a mismatch blocks the brief.
- **Re-reads the live notice** (REST table) to confirm a UPC match, instead of trusting the parser.
- **"Cheese ≠ pepperoni"**: a quote can be copied from the notice and still describe another variety — every product word of our item must appear in the notice. Live run: blocked, re-judged, ruled *not ours*.
- **Lot codes the model reads must appear verbatim** in `code_info`; invented lots are dropped.
- **Brief sentences are checked** against the evidence they cite — it caught a draft that invented stores in Kentucky and Ohio.
- **Human decisions change the next run** through H-MEM, and the brief cites them.

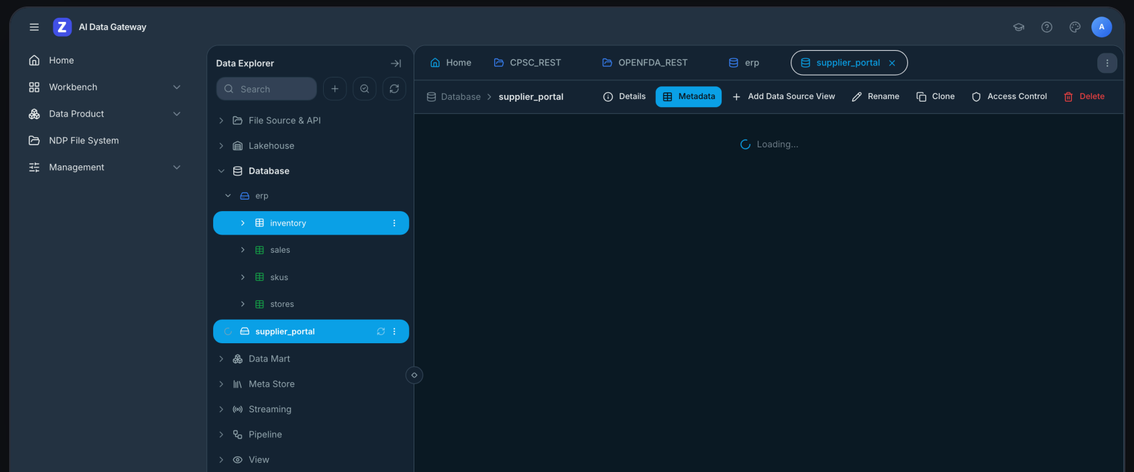
7 · PRODUCT



Brief: verdict, why (cited), actions, caveats, evidence with SQL ids — and the agent trace that produced it.



Live viewer: the four agents' events as they happen, from the trace file.



8 · RESULTS

EVAL_HEADLINE

EVAL_TABLE_HTML

Real Investigator ⇄ Auditor loop on every candidate pair from 517 recalls (sample mode, DuckDB stand-in). A pair is "covered" if the SKU was generated from that recall or its UPC is printed in the notice. Retailer data is synthetic.

9 · NEXT

From demo to a safety desk.

- More feeds through Zetaris: USDA FSIS, NHTSA equipment (car seats), Health Canada
- Lot capture from receiving scans instead of the item master
- Customer notification through the loyalty system — counts today, messages tomorrow
- Hosted Nemotron through the Meterless runtime for shared deployments
- Markovian engine for daily long-running sweeps
- Review queue UI with one-click decide → H-MEM

Limits today: synthetic retailer data; lexical Auditor checks; CPSC rarely prints UPCs; judge runs on a local Mac.